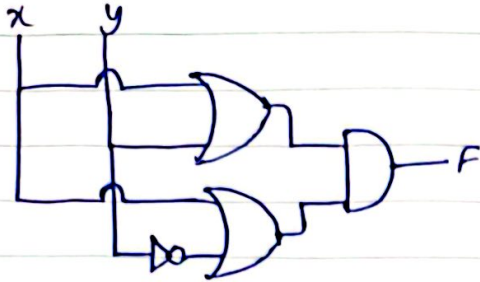


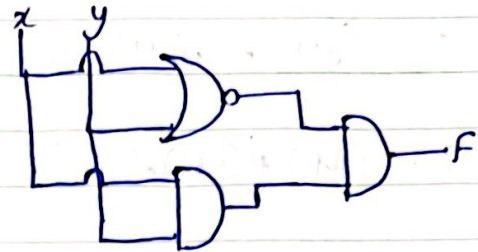
Tutorial 03

(1)

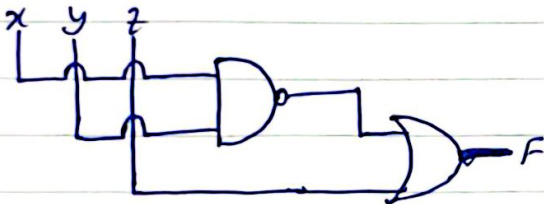
$$(1) F = (x+y)(x+\bar{y})$$



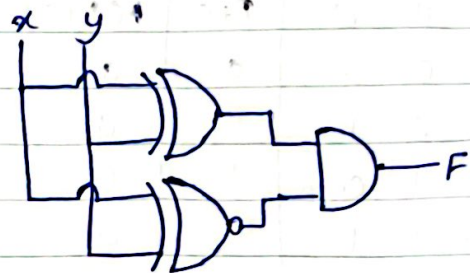
$$(2) F = \overline{(x+y)}(x \cdot y)$$



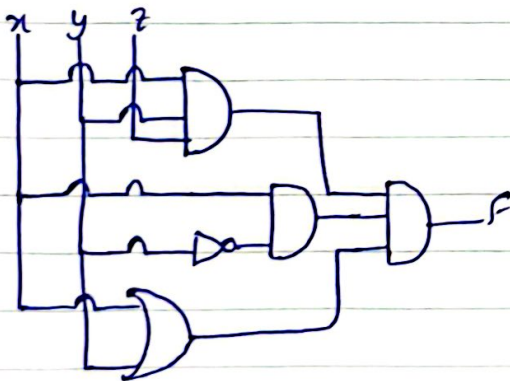
$$(3) F = \overline{x \cdot y} + z$$



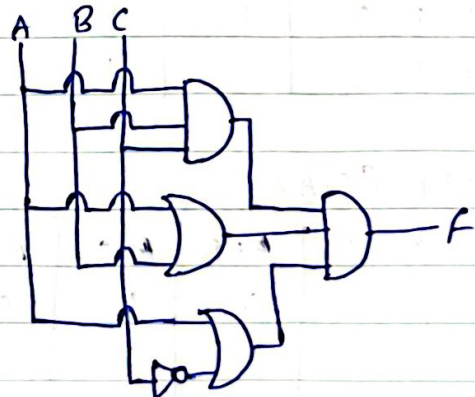
$$(4) F = (x \oplus y)(\overline{x \oplus y})$$



$$(5) F = (xyz)(x\bar{y})(x+y)$$



$$(6) F = (ABC)(A+B)(A+\bar{C})$$



(2)

$$(a) Z = (\overline{A} \cdot (A+B)) + (A+B)$$

$$(d) Y = \overline{A+B}$$

~~$$(b) Y = \overline{A+B}$$~~

$$(e) Q = (\overline{A+B}) + (B \cdot C)$$

$$(b) F = (\overline{A+B}) \cdot \overline{C} \cdot (C+D)$$

$$(c) Q = (A \cdot B) + ((B+C) \cdot (B \cdot C))$$

(3)

(3)

$$(1) A(B+C)$$

A	B	C	B+C	ACB+C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	1	0
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

$$(2) (A+B)(\overline{C \cdot A})$$

A	B	C	A+B	$\overline{A \cdot C}$	$(A+B)(\overline{C \cdot A})$
0	0	0	0	1	0
0	0	1	0	1	0
0	1	0	1	1	1
0	1	1	1	1	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	1	0	0
1	1	1	1	0	0

(3) $A\bar{B}\bar{C} + \bar{A}BC$

A	B	C	$\bar{A}$	$\bar{B}$	$\bar{C}$	$A\bar{B}\bar{C}$	$\bar{A}BC$	$A\bar{B}\bar{C} + \bar{A}BC$
0	0	0	1	1	1	0	0	0
0	0	1	1	1	0	0	0	0
0	1	0	1	0	1	0	0	0
0	1	1	1	0	0	0	1	1
1	0	0	0	1	1	1	0	1
1	0	1	0	1	0	0	0	0
1	1	0	0	0	1	0	0	0
1	1	1	0	0	0	0	0	0